Mingli Sun

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1.Education

MFA in Arts and Design, Advanced Environmental Design *Sep 2022 - Jun 2025*
GPA: 4.25/5; College of Design and Innovation (D&I), **Tongji University**, Shanghai

BEng in Industrial Design *Sep 2016 - Jun 2020*
GPA: 3.69/4; Mechanical and Electrical Engineering School, **Beijing Information Science & Technology University (BISTU)**, Beijing

2.Academic Experience

Research Assistant *Jun 2025 – Jun 2026, 1 year*
College of Architecture and Urban Planning (CAUP), Tongji University

- ***Led*** *on-site technical verification and optimization for a National Key R&D Program-funded project, with a peer-reviewed Chinese journal publication.*
- ***Mentored*** *2 undergraduate research initiatives, with a peer-reviewed conference publication.*

Research Intern *Nov 2024 – Jan 2025, 3 months*
Institute of Future Human Habitats, Tsinghua Shenzhen International Graduate School

- ***Developed*** *an end-to-end digital workflow integrating structural design and performance analysis, with a peer-reviewed conference publication.*

Technical Instructor *Feb 2023 – July 2023, 6 months*
FabLab O Shanghai (Affiliated with the MIT FabLab Global Network)

- *Provided technical guidance for p5.js in STEAM courses and contributed to the AI for Kids curriculum directed by Prof. Kostas Terzidis.*

3.Publications

- **Sun, M.** & Chai, H. **(2026)**. *Intelligent Robotic Sanding System for Building Exterior Walls Integrated with Residential Building Manufacturing Machines*. *Housing Science*, 2026(6), 80–84. https://doi.org/10.3969/j.issn.1002-0454.2026.06.011 ***(Chinese journal article)***
- **Sun, M.**, Zhou, Y., Liu, Y., Zhu, Y., Yang, Q., Liu, Y., & Chai, H. **(2026).** *Graph-Based Generative Design of Mortise-and-Tenon Joints in Timber Structures. Proceedings of the 8th International Conference on Computational Design and Robotic Fabrication (CDRF2026)*. ***(Accepted; EI)***
- **Sun, M.**, Chen, J., & Terzidis, K. **(2025).** *Text-guided Fine-tuning Diffusion Model Enhanced the Spatial Envision of "Modular" Structures: Personalized Diffusion Model in the Conceptual Design of Architectural Structures. Proceedings of the 30th CAADRIA Conference: Architectural Informatics (CAADRIA 2025), Vol. 1, 49-58.* https://doi.org/10.52842/conf.caadria.2025.1.049 ***(EI)***
- **Sun, M.**, Terzidis, K., & Chen, J. **(2024).** *Facilitation or Inhibition: The Influence of Generative Artificial Intelligence on Design Reasoning for Modular Structures. Proceedings of the IASS 2024 Symposium: Redefining the Art of Structural Design*, 109. https://doi.org/10.3929/ethz-b-000728788 ***(EI)***
- **Sun, M.**, Buš, P., Chen, J., & Sun, K. **(2024).** *Co-Intelligent Design Toward Modular Structures. Proceedings of the 29th CAADRIA Conference: Accelerated Design (CAADRIA 2024)*, *Vol.3, 171–180.* https://doi.org/10.52842/conf.caadria.2024.3.171 ***(EI)***

4.Research Experience

3.1 Architectural Design

Graph Learning-Based Generative Design of Mortise-and-Tenon Timber Joints
Sep 2025 – Jun 2026

Fields: *Artificial Neural Networks, Algorithm Design, Timber Architecture*
Technology: *Graph Neural Networks (GNNs), Python, PyTorch, PyG, Rhino - Grasshopper*

Leveraging GNNs to capture the geometric, topological, and mechanical features of mortise-and-tenon joints, this project transforms the experienced-based craftsmanship into a digital and intelligent-driven paradigm.

1. *Participated in collaborative discussions to establish the research framework, construct the joint database with 200+ samples, and define graph representations.*
2. ***Developed*** *a data processing procedure to convert 3D joint samples into graph data, conducting data inspection and augmentation expanding the dataset to 1,500+;* ***formulated*** *evaluation metrics and calculation approaches for joint performance analysis;* ***developed*** *a procedure to generate model inputs based on user-defined assembly sequences and installation directions.*
3. *Contributed to model testing and optimization, and* ***primarily authored*** *the manuscript published by CDRF2026.*

Generative AI Images for Early Conceptual Design: A Creativity Support Index (CSI) Study
Feb 2024 – Jun 2025

Fields: *Generative AI, Human-AI Interaction, Design Ethics*
Technology: *Stable Diffusion (SD), LoRA, Python, PyTorch*

This study investigates how generative AI images influence early conceptual design, employing the Creativity Support Index (CSI) and verbal protocol analysis to study human–AI interaction.

1. ***Proposed and led*** *the research project, including defining research questions, planning technical route, and designing experimental protocols.*
2. ***Developed*** *a customized diffusion model via LoRA fine-tuning, deeply participating in dataset preparation, training configuration, model evaluation.*
3. ***Designed*** *a user-centric empirical study integrating verbal protocol analysis and quantitative metrics.* ***Co-authored*** *a peer-reviewed conference paper.*

An Imitation Learning-Based Design Agent for Modular Assembly-Oriented Structures
Jun 2023 – Sep 2023

Fields: *Human-AI Interaction, Imitation Learning, Collaborative Design*
Technology: *Generative Adversarial Imitation Learning (GAIL), Unity3D, ML-Agents Toolkit*

Trains an agent through Generative Adversarial Imitation Learning (GAIL) and Proximal Policy Optimization (PPO) in Unity3D with ML-Agents Toolkit, using human demonstration trajectories to learn design reasoning strategies for modular assembly-oriented structures.

1. ***Developed*** *a Unity3D-based interactive environment for agent training, including environment and object scripting, parameter configuration, and interaction mechanism implementation.*
2. *Implemented agent training using GAIL within the ML-Agents Toolkit, including demonstration collection, parameter tuning, iterative training and evaluation.*
3. *Refined AI-generated outcomes considering manufacturing and assembly requirements. Co-authored a peer-reviewed conference publication.*

3.2 Construction Engineering

Intelligent Robotic System for High-Rise Building Façade Finishing *Jul 2025 – Jun 2026*

Funded by National Key R&D Program of China* | *Fields: *Construction Robotics, Algorithm Design*

Develops an intelligent robotic system integrated with high-rise building construction platforms for façade spraying and grinding operations to improve construction safety and efficiency.

1. ***Led*** *on-site experiments, parametrically modeled operation and environment,* ***established*** *evaluation metrics for operational performance. (contributes to the later-stage validation)*
2. ***Published*** *a peer-reviewed journal article in Chinese and co-invented a patent.*

3.3 Arts and Design

Uban Park Digital Art Driven by Eco-Acoustic Index Visualization *Sep. 2023 – Dec. 2023*

Fields: *Data Visualization, Communication Design* **|** ***Technology:*** *Tableau, TouchDesigner, Arduino*

Designed a digital art experience driven by real-time eco-acoustic index visualization to support ecological education and public awareness of urban bird habitats.

1. *Collaborated with environmental science researchers in interdisciplinary workshops and field research, and* ***developed*** *visual encoding strategies for eco-acoustic data.*
2. ***Independently designed and implemented*** *the installation through TouchDesigner-Arduino integration. The project outcome was published in a peer-reviewed design conference CUMULUS.*

5. Honors, Awards & Grants

Shanghai Outstanding Graduate Award (top 5%)	*Shanghai Municipal Education Commission*	*2025*
Xiangcheng High-Tech Talent Scholarship (endowed, top ~1%)	*Tongji University*	*2024*
International Academic Conference Travel Grant	*Tongji University*	*2024*
International and Hong Kong, Macao, and Taiwan Exchange Grant	*Tongji University*	*2023*
Beijing Outstanding Graduate Award (top 5%)	*Beijing Municipal Education Commission*	*2020*

6. Professional Skills

- ***Machine Learning Methods:*** *Graph Neural Networks, Imitation Learning, Diffusion Models*
- ***Programming Languages & Frameworks:*** *Python, C#, PyTorch, PyTorch Geometric*
- ***Computational Geometry:*** *Algorithm Development, Parametric Modeling (Rhino, Grasshopper)*
- ***Digital Fabrication & Construction:*** *3D Printing, CNC Fabrication, Construction Robotics*
- ***Empirical Research Methods:*** *Field Research, Interviews, Persona Development, Think-Aloud*

****Languages:*** *Mandarin Chinese (Native), English (Professional Working Proficiency)*

7. References

Prof. Kostas Terzidis
Professor, College of Design and Innovation (D&I), Tongji University, Shanghai, China
Email: kostas@tongji.edu.cn
Relationship: Master's Supervisor

Prof. Hua Chai
Associate Professor, College of Architecture and Urban Planning (CAUP), Tongji University, Shanghai, China
Email: chaihua@tongji.edu.cn
Relationship: Research Advisor